

Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – X

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Date of Assignment: 03-09-2026	Date/Time of Submission:03-09-2026

A) Connect and load data into Tableau Prep

Step 1: Create the Dataset Files. The datasets can be created using Python

Step 2: Generate the Term 1 Dataset. Use the following code :

```
term1_data = {  
    'Student_ID': ['S101', 'S101', 'S102', 'S102', 'S103', 'S103',  
                  'S104', 'S104', 'S105', 'S105', 'S106', 'S106',  
                  'S107', 'S107', 'S108', 'S108'],  
    'Student_Name': ['Aarav Sharma ', 'Aarav Sharma', 'priya VERMA',  
                    'priya VERMA', 'Rohan Gupta', 'Rohan Gupta',  
                    'Sneha Patel', 'Sneha Patel', 'Aditya Rao ']
```

'Aditya Rao ', 'Ananya Singh', 'Ananya Singh',

'Rahul Nair', 'Rahul Nair', 'Kavya Reddy',

'Kavya Reddy'] ,

'Department': ['Computer Science', 'Comp Sci', 'Electronics', 'ECE',

'CS', 'CSE', 'Information Tech', 'IT',

'Mechanical', 'Mech Engg', 'Computer Science', 'CS',

'ECE', 'Electronics Engg', 'IT', 'Info Tech'] ,

'Assessment_Type': ['Midterm', 'Final', 'Midterm', 'Final',

'Midterm', 'Final', 'Midterm', 'Final',

'Midterm', 'Final', 'Midterm', 'Final',

'Midterm', 'Final', 'Midterm', 'Final'] ,

'Exam_Date': ['2025/09/15', '2025/11/20', '15-09-2025',

'20-11-2025', '2025.09.15', '2025.11.20',

'2025/09/15', '2025/11/20', '2025/09/15',

'2025/11/20', '2025/09/15', '2025/11/20',

'2025/09/15', '2025/11/20', '2025/09/15',

'2025/11/20'] ,

'Subject_Code': ['CS101', 'CS101', 'EC101', 'EC101',

'CS101', 'CS101', 'IT101', 'IT101',

'ME101', 'ME101', 'CS101', 'CS101',

'EC101', 'EC101', 'IT101', 'IT101'] ,

'Marks_Obtained': [85, 92, 70, 78, 45, 52, 90, 96,

```

        35, 40, 88, 94, 65, 72, 92, 98],
'Max_Marks': [100, 100, 100, 100, 100, 100, 100, 100, 100,
              100, 100, 100, 100, 100, 100, 100, 100],
'Enrollment_Status': ['Active', 'Active', 'Active', 'Active',
                       'Active', 'Active', 'Active', 'Active',
                       'Dropped', 'Dropped', 'Active', 'Active',
                       'Active', 'Active', 'Active', 'Active']
}

```

Step 3: Produce the Term 2 Dataset. Execute the following script:

```

term2_data = {
    'Std_ID': ['S101', 'S101', 'S102', 'S102', 'S103', 'S103',
              'S104', 'S104', 'S106', 'S106', 'S107', 'S107',
              'S108', 'S108'],
    'Full_Name': ['Aarav Sharma', 'Aarav Sharma', 'Priya Verma',
                  'Priya Verma', 'Rohan Gupta', 'Rohan Gupta',
                  'Sneha Patel', 'Sneha Patel', 'Ananya Singh',
                  'Ananya Singh', 'Rahul Nair', 'Rahul Nair',
                  'Kavya Reddy', 'Kavya Reddy'],
    'Dept': ['Computer Science', 'CS', 'ECE', 'Electronics',
             'CSE', 'CS', 'IT', 'Information Tech',

```

'CSE', 'Computer Science', 'ECE', 'Electronics',

'IT', 'IT']],

'Exam_Type': ['Midterm', 'Final', 'Midterm', 'Final',

'Midterm', 'Final', 'Midterm', 'Final',

'Midterm', 'Final', 'Midterm', 'Final',

'Midterm', 'Final']],

'Assessment_Date': ['2026/02/10', '2026/04/25',

'2026/02/10', '2026/04/25',

'2026/02/10', '2026/04/25',

'2026/02/10', '2026/04/25',

'2026/02/10', '2026/04/25',

'2026/02/10', '2026/04/25',

'2026/02/10', '2026/04/25']],

'Course_Code': ['CS201', 'CS201', 'EC201', 'EC201',

'CS201', 'CS201', 'IT201', 'IT201',

'CS201', 'CS201', 'EC201', 'EC201',

'IT201', 'IT201']],

'Score': [88, 95, 75, 82, 50, 60, 94, 98,

90, 96, 70, 75, 95, 100]],

'Total_Possible': [100, 100, 100, 100, 100, 100, 100,

100, 100, 100, 100, 100, 100, 100]],

'Enrollment_Status': ['Active', 'Active', 'Active', 'Active',

```
        'Active', 'Active', 'Active', 'Active',  
        'Active', 'Active', 'Active', 'Active',  
        'Active', 'Active']  
}
```

The Term 2 dataset intentionally incorporates variations in column naming, including Std_ID, Full_Name, Dept, and Score, to facilitate later integration steps.

Step 4: Define the Course Master Dataset. Enter the code below:

```
course_data = {
```

```
'Subject_Code': ['CS101', 'CS201', 'EC101', 'EC201',  
                 'IT101', 'IT101', 'ME101'],  
  'Subject_Title': ['Data Structures', 'Algorithms',  
                   'Digital Logic', 'Signals & Systems',  
                   'Database Systems', 'Cloud Computing',  
                   'Thermodynamics'],  
  'Credit_Hours': [4, 4, 3, 4, 4, 4, 3],  
  'Passing_Marks': [40, 40, 40, 40, 40, 40, 40]  
}
```

This master file captures essential course metrics including credit weighting and threshold marks.

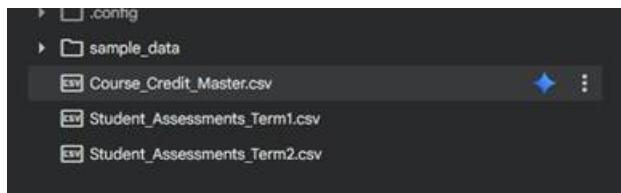
Step 5: Initialize DataFrames. Execute:

```
df_t1 = pd.DataFrame(term1_data) df_t2 =  
pd.DataFrame(term2_data) df_c =  
pd.DataFrame(course_data)
```

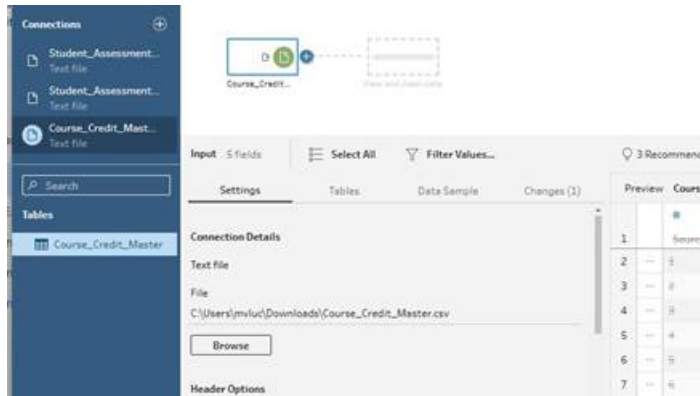
Step 6: Export Data as CSV Files. Enter:

```
df_t1.to_csv('Student_Assessments_Term1.csv', index=False)  
df_t2.to_csv('Student_Assessments_Term2.csv', index=False) df_c.to_csv('Course_Credit_Master.csv',  
index=False)
```

Follow with: `print("All 3 CSV datasets for Lesson 10 created successfully.")` The generated CSV assets will reside in your active project directory.



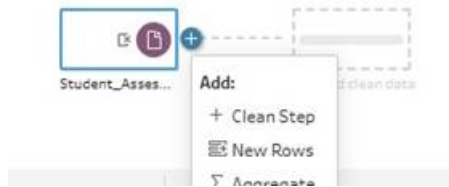
Successful completion of this practical relies on establishing connections for all three dataset components in Tableau Prep.



B) Clean and profile data

Step 1 : Insert a Cleaning Component.

1. Pick the Term 1 source node.
2. Activate the + symbol.
3. Select the **Clean Step** function.



4. A new cleaning phase is integrated into the flow.

Step 2 : Analyze the Profile Interface.

1. Enter the Clean Step view.
2. Review the Profile Pane layout.
3. Audit the entries in every column.
4. Locate trailing or leading whitespace in names.

5. Spot anomalies in letter casing.
6. Verify varying departmental labels.
7. Detects inconsistent date representation styles.
8. Scan for records of inactive students.

Tableau Prep facilitates the investigation of record dispersion and metadata health through its profile summary tools.

Step 1: Select the Department Field.

1. Access the active **Clean Step** component.
2. Find the Department column header.
3. Click the **More Options (...)** icon.
4. Navigate to **Group Values**.
5. Pick the **Manual Selection** method.

Step 2: Consolidate Computer Science Entries. Choose the variant strings: Computer

Science, Comp Sci, CS, and CSE. Assign them to a unified category titled: Computer

Science

Student_ID	Student_Name	Department	Assessment_Type ↑	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment
S101	Aarav Sharma	Comp Sci	Final	2025/11/20	CS101	92	100	Active
S106	Ananya Singh	Comp Sci	Final	2025/11/20	CS101	94	100	Active
S103	Rohan Gupta	Comp Sci	Final	2025.11.20	CS101	52	100	Active
S107	Rahul Nair	Electronics Engg	Final	2025/11/20	EC101	72	100	Active

Step 3: Consolidate Electronics Entries. Select the labels: Electronics, ECE, and

Electronics Engg. Standardize this group under the name: Electronics

Student_ID	Student_Name	Department	Assessment_Type ↑	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment
S102	priya VERMA	Electronics Engg	Final	20-11-2025	EC101	78	100	Active
S107	Rahul Nair	Electronics Engg	Final	2025/11/20	EC101	72	100	Active
S107	Rahul Nair	Electronics Engg	Midterm	2025/09/15	EC101	65	100	Active
S102	priya VERMA	Electronics Engg	Midterm	15-09-2025	EC101	70	100	Active

Step 4: Consolidate Information Technology Entries. Identify the values: Information

Tech, IT, and Info Tech. Map these to the primary designation: Information Technology

Student_ID	Student_Name	Department	Assessment_Type ↑	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment
S108	Kavya Reddy	Information Tech	Final	2025/11/20	IT101	98	100	Active
S104	Sneha Patel	Information Tech	Final	2025/11/20	IT101	96	100	Active
S104	Sneha Patel	Information Tech	Midterm	2025/09/15	IT101	90	100	Active
S108	Kavya Reddy	Information Tech	Midterm	2025/09/15	IT101	92	100	Active

Step 5: Consolidate Mechanical Engineering Entries. Select: Mechanical and Mech Engg. Rename the resulting group as: Mechanical Engineering. This process ensures data integrity by reconciling disparate naming conventions into uniform academic departments.

Step 1: Establish a Data Union.

1. Highlight the Term 1 input.
2. Pull the Term 2 asset onto it.
3. Pick the **Union** capability.
4. Tableau Prep identifies aligned and divergent headers.

Step 2: Reconcile Student Identifier Fields. Link: Student_ID → Std_ID

Step 3: Reconcile Student Naming Fields. Link: Student_Name → Full_Name

Step 4: Reconcile Academic Department Fields. Link: Department → Dept

Step 5: Reconcile Examination Category Fields. Link: Assessment_Type →
Exam_Type

Step 6: Reconcile Chronological Fields. Link: Exam_Date → Assessment_Date

Step 7: Reconcile Course Designation Fields. Link: Subject_Code → Course_Code

Step 8: Reconcile Performance Score Fields. Link: Marks_Obtained → Score

Step 9: Reconcile Maximum Threshold Fields. Link: Max_Marks → Total_Possible

Integrating these files through a Union allows for a comprehensive overview while correcting naming inconsistencies across the datasets.

The screenshot displays a data integration workflow and a detailed view of a Union operation. At the top, a visual flow shows a data source 'Student_Assess...' connected to a 'Union 1' block, which then connects to a 'Join 3' block. Below this, the 'Union 1' configuration window is open, showing 11 fields and 44 rows. The 'Settings' tab is active, displaying 'Inputs' (Clean 1, Student_Assessments_Term2-4, Clean 3) and 'Resulting Fields' (2 mismatching fields from 11 resulting fields). The 'Mismatched Fields' section lists 'Department' and 'Dept'. The 'Union Results' tab is also visible, showing a comparison between 'Department' and 'Dept' with a list of mismatched values.

Department	Dept
Comp Sci	null
null	IT
Electronics Engg	null
null	IT

E) Perform aggregations in Prep

Step 1: Insert a Cleaning Component Post-Union.

1. Locate the Union node and select the + icon.
2. Choose the **Clean Step** option.
3. This appends a new processing phase to the unioned data.

Step 2: Define the Term Category Field.

1. Enter the newly added Clean Step.
2. Access the **More Options (...)** menu.
3. Navigate to **Create a Calculated Field**.
4. Label the field as: Term
5. Apply the following logic:

IF CONTAINS([Table Names], "Term1") THEN "Term 1"

ELSEIF CONTAINS([Table Names], "Term2") THEN "Term 2"

ELSE "Other"

END

6. Select **Save** to finalize.

This logic effectively sorts records into their respective academic periods.

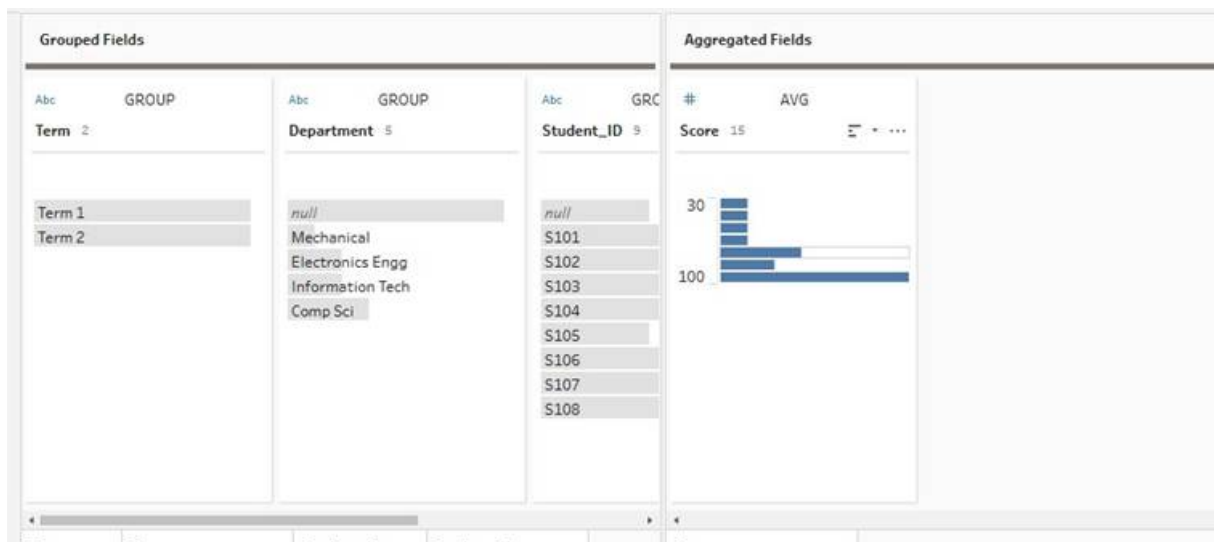
Step 3: Add an Aggregate Component.

1. Hover over the active Clean Step.
2. Select the + symbol.
3. Opt for the **Aggregate** function.
4. The aggregation workspace becomes available.

The interface displays panes for **Grouped Fields** and **Aggregated Fields**. **Step 4:** Specify Grouping

Dimensions. Map the following dimensions to **Grouped Fields**: Student_ID Student_Name Department

Term



F) Apply filters and transformations

Step 1: Access the Processing Component.

1. Select the Clean Step following the aggregation node.

2. Audit the resulting summarized fields.
3. Confirm existence of the following parameters:

Student_ID Student_Name Department Term Total_Marks_Obtained Average_Score Total_Max_Marks_{and}

Assessments_Taken

Step 2: Filter Out Inactive Records.

1. Identify the Enrollment_Status column.
2. Choose the **Dropped** category value.
3. Right-click and select **Exclude**.
4. Alternatively, apply a filter to retain only **Active** scholars.

This procedure isolates relevant data by purging withdrawn students from the flow.

Step 3: Compute Term Performance Percentage.

1. Select **Create Calculated Field**.
2. Designate the title as: Term_Percentage
3. Input the calculation logic:

$$\text{ROUND}\left(\left(\left[\text{Total_Marks_Obtained}\right] / \left[\text{Total_Max_Marks}\right]\right) * 100, 2\right)$$

4. Finalize with **Save**.

Note: If original field names persist, employ: $\text{ROUND}\left(\left(\left[\text{Score}\right] / \left[\text{Total_Possible}\right]\right) * 100, 2\right)$

Step 4: Determine Alpha Grade Categories.

1. Navigate to **Create a Calculated Field**.
2. Label as: Letter_Grade
3. Apply the conditional logic:

```
IF [Term_Percentage] >= 85 THEN "A (Distinction)"  
  
ELSEIF [Term_Percentage] >= 75 THEN "B (First Class)"  
  
ELSEIF [Term_Percentage] >= 60 THEN "C (Second Class)"  
  
ELSEIF [Term_Percentage] >= 40 THEN "D (Pass Class)"  
  
ELSE "F (Fail / Re-appear)"  
  
END
```

4. Click **Save**. This classifies academic achievement into uniform grade brackets.

Step 5: Establish Academic Status Field.

1. Initiate **Create Calculated Field**. 2. Label as:

Academic_Standing

3. Apply logic:

```
IF [Term_Percentage] >= 60 THEN "Satisfactory"
```

ELSE "Academic Probation (<60%)"

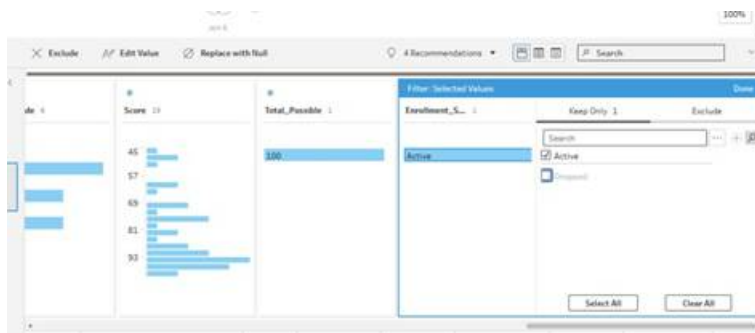
END

4. Select **Save**.

Step 6: Validate Processed Metadata.

1. Inspect the Term_Percentage distribution for numeric validity.
2. Audit the Letter_Grade column for accurate categorization.
3. Verify correct designation in the Academic_Standing field.

Reviewing these profile summaries ensures structural integrity following the transformation phase.



G) Create and automate data preparation flows

Step 1: Insert an Export Component.

1. Locate the final processing node and select the + symbol.
2. Pick the **Output** functionality.
3. Determine the destination for your file.

Step 2: Determine the Export Format. Opt for either: Comma Separated Values (.csv) or Tableau

Data Extract (.hyper)

RESULT :- The educational datasets were integrated and transformed within Tableau Prep. Through rigorous cleaning and aggregation, inactive records were filtered while performance metrics like grades and standing were derived and exported successfully.